Chapter 1 Checklist

Functions and Their Graphs

Ron Larson and Robert P. Hostetler, *Precalculus with Limits*, 2nd ed., p. 1. Tick each box as you go.

Work through the four parts in order: read each section, practise the matching Duolingo units, work the book-only topics that Duolingo does not cover, then finish the chapter. Duolingo units are grouped by lane (a grade or a topic in the app) and listed in the order the app shows them, so you can follow one lane from top to bottom. **full** = the unit practises the section's main skill, **partial** = part of it, **prereq** = a warm-up from an earlier grade.

In this chapter: Grade 9 carries the basics (functions, lines, transformations), grade 11 repeats them at precalculus depth, and the Algebraic Graphing topic is a condensed walk through the whole chapter. Follow grade 11 if you only have time for one lane.

1. Read the textbook sections

- ☐ 1.1 Rectangular Coordinates p. 2
- ☐ 1.2 Graphs of Equations p. 13
- ☐ 1.3 Linear Equations in Two Variables p. 24
- ☐ 1.4 Functions p. 39
- ☐ 1.5 Analyzing Graphs of Functions p. 54
- ☐ 1.6 A Library of Parent Functions p. 66
- ☐ 1.7 Transformations of Functions p. 73
- ☐ 1.8 Combinations of Functions: Composite Functions p. 83
- ☐ 1.9 Inverse Functions p. 92
- ☐ 1.10 Mathematical Modeling and Variation p. 102

2. Practise on Duolingo

Pick one lane and work down it; the grade lanes go deepest. The note after each unit gives its rating and the textbook sections it serves.

Grade 6

- ☐ Plot points in all quadrants *prereq, 1.1* unit 33

Grade 7

- ☐ Proportionality in equations *prereq, 1.10* unit 7

Grade 8

- ☐ Intro to slope *prereq, 1.3* unit 21
- ☐ Slope formula *prereq, 1.3* unit 22
- ☐ Slope-intercept form *prereq, 1.3* unit 24

- ☐ Equations of parallel lines *full, 1.3* unit 28
- ☐ Equations of perpendicular lines *full, 1.3* unit 29
- ☐ Intro to functions *prereq, 1.4* unit 36

Grade 9

- ☐ Slope-intercept form of linear equations *full, 1.3* unit 5
- ☐ Standard form of linear equations *full, 1.3* unit 7
- ☐ Inputs and outputs *full, 1.4* unit 17
- ☐ Graphs of functions *full, 1.5* unit 18
- ☐ Vertical intercept of functions *partial, 1.2* unit 20
- ☐ Horizontal intercept of functions *partial, 1.2* unit 21
- ☐ Average rate of change *full, 1.5* unit 23
- ☐ Linear functions *prereq, 1.6* unit 24
- ☐ Domain of a function *full, 1.4* unit 25
- ☐ Range of a function *partial, 1.4* unit 26
- ☐ Graphs of piecewise functions *full, 1.5* unit 27
- ☐ Equations of piecewise functions *full, 1.4* unit 28
- ☐ Step functions *full, 1.6* unit 30
- ☐ Absolute value functions *full, 1.6* unit 31
- ☐ Graphs of inverse functions *partial, 1.9* unit 33
- ☐ Equations of inverse functions *partial, 1.9* unit 34
- ☐ Graphs of inverse linear functions *prereq, 1.9* unit 35
- ☐ Equations of inverse linear functions *prereq, 1.9* unit 36
- ☐ Vertical translations *full, 1.7* unit 75
- ☐ Horizontal translations *full, 1.7* unit 76
- ☐ Vertical stretches *full, 1.7* unit 77
- ☐ Vertical compressions *full, 1.7* unit 78
- ☐ Horizontal stretches *full, 1.7* unit 79
- ☐ Horizontal compressions *full, 1.7* unit 80
- ☐ Scatter plots *partial, 1.10* unit 81
- ☐ Correlation *partial, 1.10* unit 82
- ☐ Linear models *partial, 1.10* unit 83
- ☐ Residuals *partial, 1.10* unit 84

Grade 10

- ☐ Reflectional symmetry *prereq, 1.2* unit 10
- ☐ Lines in point-slope form *full, 1.3* unit 45
- ☐ Lines in different forms *full, 1.3* unit 46
- ☐ Circles on the plane *full, 1.2* unit 47
- ☐ Line segment midpoint *full, 1.1* unit 50
- ☐ Distance formula *full, 1.1* unit 51

Grade 11

- ☐ Function families *full, 1.6* unit 2
- ☐ Compare function graphs *full, 1.5* unit 3
- ☐ Transform absolute value functions *full, 1.7* unit 9
- ☐ Relative maxima and minima of polynomial functions *partial, 1.5* unit 13
- ☐ Inverse function equations *full, 1.9* unit 65
- ☐ Inverse function graphs *full, 1.9* unit 66
- ☐ Combine functions *full, 1.8* unit 67
- ☐ Function composition *full, 1.8* unit 68
- ☐ Vertical translations *full, 1.7* unit 69
- ☐ Horizontal translations *full, 1.7* unit 70
- ☐ X-axis function reflections *full, 1.7* unit 71
- ☐ Y-axis function reflections *full, 1.7* unit 72
- ☐ Vertical stretches *full, 1.7* unit 73
- ☐ Vertical compressions *full, 1.7* unit 74
- ☐ Horizontal stretches *full, 1.7* unit 75
- ☐ Horizontal compressions *full, 1.7* unit 76

Grade 12

- ☐ Function composition *full, 1.8* unit 13
- ☐ Inverse functions *full, 1.9* unit 14
- ☐ Domain and range *full, 1.5* unit 17
- ☐ Graphing inverses *full, 1.9* unit 18

Geometry

- ☐ Absolute value and distance *prereq, 1.1* unit 11

Algebraic Graphing

- ☐ Slope *partial, 1.3* unit 3
- ☐ Slope-intercept lines *partial, 1.3* unit 4
- ☐ Function notation *full, 1.4* unit 8
- ☐ Piecewise-defined functions *full, 1.4* unit 9
- ☐ Step and absolute value functions *full, 1.6* unit 10
- ☐ Inverse linear functions *partial, 1.9* unit 23
- ☐ Function translations *full, 1.7* unit 24
- ☐ Vertical function scaling *full, 1.7* unit 25
- ☐ Horizontal function scaling *full, 1.7* unit 26
- ☐ Reflections of functions *full, 1.7* unit 27

Statistics

- ☐ Linear models *partial, 1.10* unit 15
- ☐ Linear model fitness *partial, 1.10* unit 16

3. Section by section

For each section: what Duolingo covers (the units are ticked off in part 2) and the book-only topics to work in the textbook. Tick each book-only topic once you have worked examples of it.

1.1 Rectangular Coordinates p. 2

Distance and midpoint are covered in grade 10. Plotting is a lower-grade prerequisite. The section's data applications are book only.

Duolingo: **Grade 6:** 33. Plot points in all quadrants (prereq)
Grade 10: 50. Line segment midpoint (full); 51. Distance formula (full)
Geometry: 11. Absolute value and distance (prereq)

Book only:

- ☐ Translating points in the plane
- ☐ Applications of the distance and midpoint formulas

1.2 Graphs of Equations p. 13

Intercepts and circles are covered; symmetry tests and point-plotting of general equations are book only.

Duolingo: **Grade 9:** 20. Vertical intercept of functions (partial); 21. Horizontal intercept of functions (partial)
Grade 10: 10. Reflectional symmetry (prereq); 47. Circles on the plane (full)

Book only:

- ☐ Symmetry tests (x-axis, y-axis, origin)
- ☐ Sketching graphs of general equations by point plotting

1.3 Linear Equations in Two Variables p. 24

Well covered. Grade 8 builds slope and parallel/perpendicular lines, grade 9 the forms of a line, grade 10 point-slope form.

Duolingo: **Grade 8:** 21. Intro to slope (prereq); 22. Slope formula (prereq); 24. Slope-intercept form (prereq); 28. Equations of parallel lines (full); 29. Equations of perpendicular lines (full)
Grade 9: 5. Slope-intercept form of linear equations (full); 7. Standard form of linear equations (full)
Grade 10: 45. Lines in point-slope form (full); 46. Lines in different forms (full)
Algebraic Graphing: 3. Slope (partial); 4. Slope-intercept lines (partial)

Book only:

- ☐ Slope as a rate of change in applications

1.4 Functions p. 39

Function definition, notation, domain, and piecewise functions are covered. The difference quotient is book only.

Duolingo: **Grade 8:** 36. Intro to functions (prereq)
Grade 9: 17. Inputs and outputs (full); 25. Domain of a function (full); 26. Range of a function (partial); 28. Equations of piecewise functions (full)
Algebraic Graphing: 8. Function notation (full); 9. Piecewise-defined functions (full)

Book only:

- ☐ The difference quotient
- ☐ Writing a function from a word problem

1.5 Analyzing Graphs of Functions p. 54

Domain and range from graphs, zeros, relative extrema, and average rate of change are covered. Even and odd functions and the vertical line test are book only.

Duolingo: **Grade 9:** 18. Graphs of functions (full); 23. Average rate of change (full); 27. Graphs of piecewise functions (full)
Grade 11: 3. Compare function graphs (full); 13. Relative maxima and minima of polynomial functions (partial)
Grade 12: 17. Domain and range (full)

Book only:

- ☐ Even and odd functions
- ☐ The Vertical Line Test

1.6 A Library of Parent Functions p. 66

Grade 11's Function families is the closest match to the parent-function library. Step and absolute value functions are drilled separately.

Duolingo: **Grade 9:** 24. Linear functions (prereq); 30. Step functions (full); 31. Absolute value functions (full)
Grade 11: 2. Function families (full)
Algebraic Graphing: 10. Step and absolute value functions (full)

Book only:

- ☐ The square root and reciprocal parent functions in detail

1.7 Transformations of Functions p. 73

Fully covered. Grade 9 drills each transformation; grade 11 repeats them with reflections added. Do grade 11 if time is short.

Duolingo: **Grade 9:** 75. Vertical translations (full); 76. Horizontal translations (full); 77. Vertical stretches (full); 78. Vertical compressions (full); 79. Horizontal stretches (full); 80. Horizontal compressions (full)
Grade 11: 9. Transform absolute value functions (full); 69. Vertical translations (full); 70. Horizontal translations (full); 71. X-axis function reflections (full); 72. Y-axis function reflections (full); 73. Vertical stretches (full); 74. Vertical compressions (full); 75. Horizontal stretches (full); 76. Horizontal compressions (full)
Algebraic Graphing: 24. Function translations (full); 25. Vertical function scaling (full); 26. Horizontal function scaling (full); 27. Reflections of functions (full)

1.8 Combinations of Functions: Composite Functions p. 83

Arithmetic combinations and composition are covered. Finding the domain of a composite and decomposing a function are lighter in the app than in the book.

Duolingo: **Grade 11:** 67. Combine functions (full); 68. Function composition (full)
Grade 12: 13. Function composition (full)

Book only:

- ☐ Domain of a composite function
- ☐ Decomposing a function into two simpler functions

1.9 Inverse Functions p. 92

Finding and graphing inverses is covered in grades 9, 11, and 12. The horizontal line test and one-to-one functions are book only.

Duolingo: **Grade 9:** 33. Graphs of inverse functions (partial); 34. Equations of inverse functions (partial); 35. Graphs of inverse linear functions (prereq); 36. Equations of inverse linear functions (prereq)
Grade 11: 65. Inverse function equations (full); 66. Inverse function graphs (full)
Grade 12: 14. Inverse functions (full); 18. Graphing inverses (full)
Algebraic Graphing: 23. Inverse linear functions (partial)

Book only:

- ☐ One-to-one functions and the Horizontal Line Test

1.10 Mathematical Modeling and Variation p. 102

Least squares and linear models are covered in grade 9 and the Statistics topic. Direct variation has a grade 7 prerequisite; inverse and joint variation are book only.

Duolingo: **Grade 7:** 7. Proportionality in equations (prereq)
Grade 9: 81. Scatter plots (partial); 82. Correlation (partial); 83. Linear models (partial); 84. Residuals (partial)
Statistics: 15. Linear models (partial); 16. Linear model fitness (partial)

Book only:

- ☐ Inverse, joint, and combined variation
- ☐ Least squares regression on the calculator

4. Finish the chapter

- ☐ Chapter Summary p. 114

- ☐ Review Exercises p. 116
- ☐ Chapter Test p. 121
- ☐ Proofs in Mathematics p. 122
- ☐ Problem Solving p. 123

Duolingo unit numbers are each unit's position in that grade's or topic's path in the Duolingo Math app, as recorded on 2026-10-07 (iOS); the app may reorder them. Textbook: Ron Larson and Robert P. Hostetler. Precalculus with Limits, 2nd ed. (Pre-AP National Student Edition) Thomson Brooks/Cole, 2009. ISBN 978-0-538-73659-6. Only titles and unit names are reproduced, for reference; no textbook or Duolingo content is included. This checklist is licensed CC BY 4.0; see the repository NOTICE for third-party material.